

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0665

Halifax 138 kV
October 01, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0665. The Interconnection Customer proposes a 26.7 MW solar photovoltaic facility interconnecting at the Halifax 138 kV substation in the Virginia area. The request was received on August 20, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 313 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$620,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
AIBONITO PDS SUBSTATION / SENANDOAH 138 kV Line	BEAR / CLOVER 345 kV	109.0	293	5.5
MECKLENBURG POER STA / PERSON 345 kV	CLOVER / BEAR 345 kV L	101.7	2,318	19.6
SHENANDOAH / WALWORTH 138 kV	CLOVER / BEAR 345 kV	100.9	297	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AIBONITO PDS SUBSTATION / SENANDOAH 138 kV Line	Reconductor line	\$13,535,000	2% (\$260,000)
MECKLENBURG POER STA / PERSON 345 kV	Reconductor line	\$33,349,000	0% (\$134,000)
WALWORTH / SHENANDOAH 138 kV	Reconductor line	\$11,760,000	2% (\$226,000)
	Total allocated to project		\$620,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.